

A New Electronic Voltage Transformer Applicable to AMI Smart Meters

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Abstract—Voltage transformers (VTs) used in the distribution-level smart meters in advanced metering infrastructure (AMI) systems are mainly composed of iron cores and windings, and are insulated with epoxy resin. Therefore, they are bulky and heavy. In addition, the complex winding structure easily leads to uneven epoxy resin grouting, so partial discharge may occur between layers in VTs, which may eventually lead to explosion accidents.

This study proposes a new electronic voltage transformer (EVT), whose secondary voltage output not only meets the accuracy required by the AMI smart meter but also needs to be used to provide the power required by the AMI smart meter. The new EVT is based on the principle of resistive voltage dividers. Moreover, a push-pull power amplifier is combined to realize high power output and impedance matching. Besides, the new EVT adopts a modular design. For different distribution voltage applications, the voltage measurement requirements of 380 V/114 V, 11.4 kV/114 V or 22.8 kV/114 V can be realized by changing the cascade module. The experimental results show that the new EVT for the 380 V/114 V application not only successfully drives the AMI smart meter, but also meets the Class 0.1 according to IEC standard 60044-7.

Keywords—electronic voltage transformer (EVT), voltage transformer (VT), resistive voltage divider, smart meter, advanced metering infrastructure (AMI)

I. INTRODUCTION

In addition to supplying the accurate voltage parameters required by the metering device, the voltage transformer (VT) also immediately supplies the instantaneous waveform required by the protective system. The VT converts the high voltages in the power system into the working voltage (220 V or 110 V) allowed by meters, Distribution boards, and protective devices according to fixed ratios. The VT structure of the traditional electromagnetic induction technology is relatively complicated. Moreover, the whole structure is often encapsulated with epoxy resin to achieve the effect of high-voltage insulation. Figure 1 shows a distribution level (22 kV) traditional VT and its explosion accident. When the epoxy resin grouting technology is immature, the process of encapsulation and molding may easily cause uneven grouting inside the EVT, resulting in small holes. Therefore, there is a possibility of partial discharge (PD) between the layers. The PD may directly lead to damage to the insulating medium, reduce equipment life [1, 2], and even cause explosion accidents. In addition, in the process of assembling the winding on the iron core of the traditional VT, the winding

wire may be damaged due to improper engineering, which may also cause discharge and short circuit between the windings in the future. Overall, due to the structure of the iron core and epoxy resin insulation, the traditional VT is bulky and heavy. Moreover, it may have problems such as ferromagnetic resonance and magnetic saturation.



Fig. 1. A 22-kV traditional VT and its explosion accident.

Compared with the traditional VT, the EVT has no iron core structure, so there is no problem with iron core magnetic saturation and ferromagnetic resonance. Since there is no magnetic saturation problem, the EVT has a wide range of transient voltage response. Also, because there is no iron core structure, the overall structure of EVT is small and light in weight. Due to the simple structure of EVT, the epoxy resin can be evenly distributed during the grouting insulation process, and it is not easy to cause discharge accidents due to uneven grouting. Therefore, the overall performance and flexibility of EVT are better than that of traditional VT. The voltage sensing principle of EVT is mainly based on semiconductor components and voltage division technology. So the secondary voltage output is small (< 5 V), and it is easier to be digitally sampled. Because of the above advantages, EVT is suitable for automation integration in intelligent substations [3, 4]. In recent years, many studies have been devoted to the development of new EVT [5–7].

This study proposes a new modular EVT design to replace the traditional VT used in distribution level (11.4 kV and 22.8 kV) AMI smart meters. Considering that the input voltage specification required by the AMI meter needs to be between 110 V and 380 V. Therefore, this study adopts resistive voltage dividers and modular structure to realize the measurement of different voltage levels (380 V, 11.4 kV, 22.8 kV). And, the secondary voltage output of the new EVT is a fixed value of 114 V. In addition, the AMI meter is generally designed to take power from the traditional VT to drive its analog and digital circuits and systems. Therefore, the new EVT not only meets the

accuracy class required for the AMI meter but also needs to have the ability to drive the AMI meter. Therefore, a high-precision push-pull power amplifier is designed. This push-pull power can increase the power output of the new EVT and optimize the impedance matching between the new EVT and the AMI meter. So the accuracy class of the new EVT still meets Class 0.1 according to IEC standard 60044-7 [8].

II. NEW ELECTRONIC VOLTAGE TRANSFORMER

EVT sensing technologies are mainly divided into semiconductor sensing methods [9, 10] and voltage division methods [11, 12]. For semiconductor sensing methods, the electro-optical effect and piezoelectric effect have complex structures and high prices, so they are suitable for medium-voltage and high-voltage related applications. In addition, semiconductor materials are easily affected by ambient temperature, and the related compensation technology is also relatively complicated. Voltage division methods include: capacitive voltage divider, inductive voltage divider, and resistive voltage divider. The capacitive voltage divider is commonly used for medium voltage to ultra-high voltage measurements. However, it lacks the flexibility of multi-stage voltage division design, and the output power is extremely low. The inductive voltage divider also has an iron core structure, so there are the same problems as the traditional VT. In this study, the resistive voltage divider is utilized to design the new EVT, which uses high-precision resistors with extremely low temperature drift, and the voltage ratio is extremely stable. Because high-precision resistors with extremely low temperature drift are selected, the voltage ratio stability of the new EVT is extremely high. Besides, with a push-pull power amplifier circuit, the new EVT can drive the AMI smart meter. Moreover, the accuracy class of the new EVT will be designed based on IEC 60044-7 Class 0.1.

A. Resistive Voltage Divider Based EVT

In this study, a modular EVT is designed for the three-phase 380 V voltage level. Then a push-pull power amplifier is designed to increase the output energy of the EVT to drive AMI smart meter. The overall architecture of this new EVT is shown in Fig. 2. Based on resistive voltage dividers, the new EVT is composed of multiple precision resistors of the same resistance in series, and then a certain resistor in the middle is used as a tap for the output of the voltage divider.

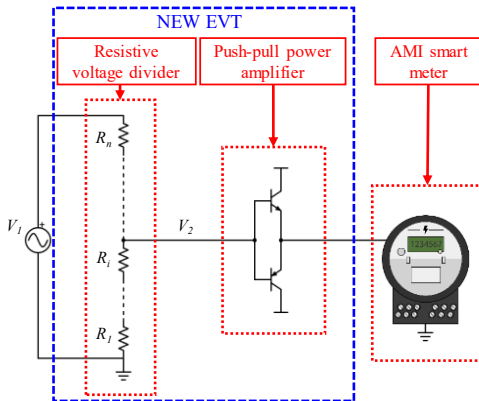


Fig. 2. New EVT architecture.

The formula corresponding to the voltage ratio of the EVT in Fig. 1 is shown in (1). In this study, ten 100-kΩ resistors are connected in series. Then, in order to satisfy the output voltage between 110 V and 380 V, the third resistor is designed as a tap output. Therefore, the ideal voltage ratio is 0.3, as shown in Fig. 3. When the primary voltage is 380 V, the secondary voltage output is 114 V. In the new EVT prototype, the precision resistor is Caddock Ultra-Stable Low TC Film Resistor with an error of 0.01 % and a temperature coefficient of 5 ppm/°C. Therefore, the absolute tolerance of the resistance value is extremely small, and characteristics of each resistor are very similar, so an accurate voltage ratio can be realized. In addition, the extremely low temperature coefficient also ensures that the voltage ratio is not easily affected by changes in ambient temperature.

$$\frac{V_2}{V_1} = \frac{\sum_{a=1}^i R_a}{\sum_{a=1}^n R_a}, \quad (1)$$

where V_1 is the high-side voltage, V_2 is the tap voltage, n is the total number of resistors, and i is the resistance number in tap position.

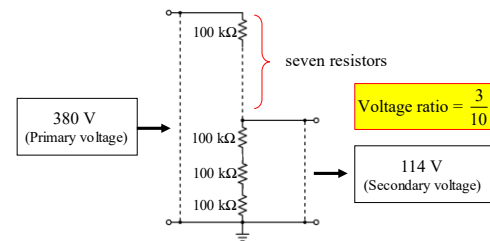


Fig. 3. Resistor scheme and voltage ratio for the 380 V/114 kV EVT.

B. Push-pull power amplifier

Although the resistive voltage divider has high accuracy, it does not have high power output capability. In fact, the AMI smart meter will generate a starting current when it is powered on, as shown in Fig. 4. This startup current must flow through the series resistor on the voltage divider. This large current may cause temperature drift at the least, or burn out the series resistor at worst.

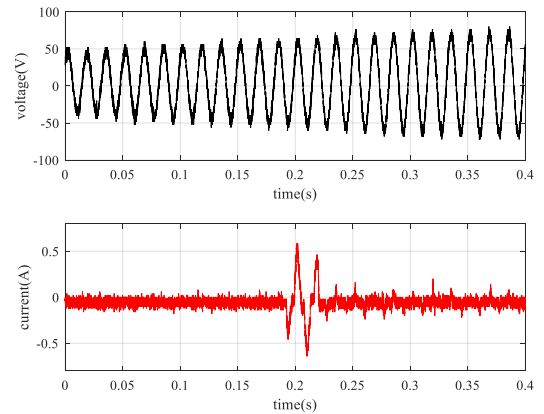


Fig. 4. The AMI smart meter's startup current when powered on.

The self-heating loss of the resistors will increase rapidly with the power of current. In this case, the accuracy level of the resistive voltage divider will degrade, and the risk of failure will also increase. In this study, a push-pull power amplifier is designed to maintain the measurement accuracy of the new EVT while ensuring sufficient output capacity to drive the AMI smart meter. The proposed push-pull power amplifier is shown in Fig. 5. Q_2 and Q_4 are transistors that provide push-pull power, while Q_1 and Q_3 are responsible for impedance matching of the equivalent series resistance of the input signal. Therefore, the push-pull power amplifier can impedance-match the resistive voltage divider (R_S) with the AMI smart meter (R_L). Finally, the new EVT maintains its measurement accuracy while driving any AMI smart meter.

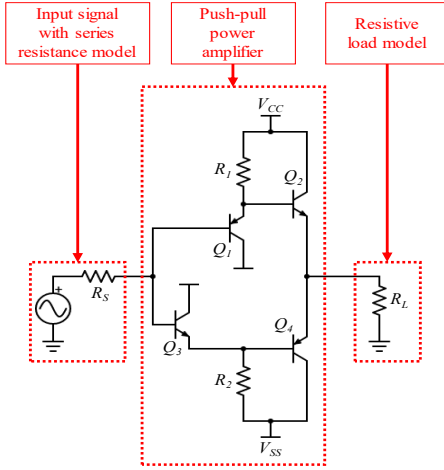


Fig. 5. Adopted push-pull power amplifier circuit.

C. Modular Architecture for new EVT

The application field of AMI smart meters includes low voltage level (380 V) and high voltage level (11.4 kV, 22.8 kV). In order to make the new EVT applicable to different voltage levels, a modular design of the EVT is proposed. Figure 6 shows the basic-layer module for the new EVT with a voltage ratio of 380V/114V. When applied to 11.4 kV, the basic-layer module is superimposed by the second-layer module, and the voltage ratio will be 11.4 kV/114 V. When applied to 22.8 kV, a third-layer module is added, and the voltage ratio will be 22.8 kV/114 V. Figure 7 is a schematic diagram of the modular architecture for the 22.8 kV/114 V EVT. Ten 100-k Ω resistors are connected in series for the basic-layer EVT module, twenty-nine 1-M Ω resistors are connected in series for the second-layer EVT module, and thirty 1-M Ω resistors are connected in series for the third-layer EVT module.



Fig. 6. Prototype of the basic-layer module for new EVT.

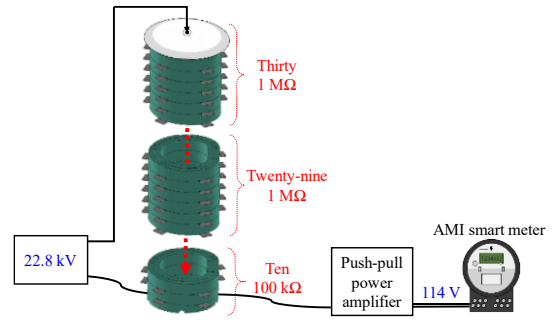


Fig. 7. Basic-layer, second-layer, and third-layer modules for new EVT in 22.8 kV/114 V applications.

III. FRAMEWORK OF THE MEASUREMENT SYSTEM

The new EVT is used to drive the AMI smart meter, and its voltage error and phase offset may be affected by this load. Therefore, it will lead to serious errors in the calculation of power/energy and electricity consumption by AMI smart meters. According to the IEC 60044-7 accuracy class 0.1 specification, this study will test whether the accuracy meets the limit values listed in Table I when the testing voltage is 80 % to 120 % of the rated voltage. In Table I, ε_u and $\Delta\phi$ are defined as (2) and (3).

TABLE I RATIO ERROR AND PHASE ERROR LIMITS FOR IEC 60044-7 ACCURACY CLASS 0.1

Class	Voltage (ratio) error ε_u (%)	Phase displacement $\Delta\phi$ (°)
0.1	0.1	5

$$\varepsilon_u = \frac{V_{DUT} - V_{ref}}{V_{ref}} \times 100\% \quad (2)$$

where

ε_u : the percentage voltage (ratio) error;

V_{DUT} : the voltage RMS value measured by the new EVT, and

$$V_{DUT} = V_{EVT(output)} \times Ratio_{EVT};$$

$V_{EVT(output)}$: the secondary-side voltage of the EVT;

$Ratio_{EVT}$: the ratio of the EVT primary-side voltage to the EVT secondary-side voltage;

V_{ref} : the voltage RMS value measured by the reference VT.

$$\Delta\phi = phase_{DUT} - phase_{ref} \quad (3)$$

where

$\Delta\phi$: the displacement;

$phase_{DUT}$: the voltage phase measured by the new EVT;

$phase_{ref}$: the voltage phase measured by the reference VT.

A measurement system for testing the new EVT is built up, as shown in Fig. 8. An AC power supply with a rated voltage of 500 V was used to generate a testing voltage of 380 V, which was simultaneously sent to the new EVT and the reference VT. Then the output signals of these two devices are digitally sampled synchronously by a NI DAQ system (NI USB-6356 BNC), and finally, the RMS error and phase displacement of the new EVT are calculated. Because the output of the new EVT is

114 V which is much higher than the input voltage limit of the NI DAQ system (± 10 V_{DC}), this voltage must be stepped down by another reference VT. The above two reference VTs are the same voltage differential probes (HIOKI P9000-01).

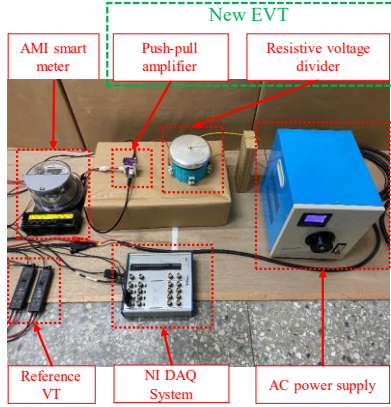


Fig. 8. Measurement system for new EVT.

IV. MEASUREMENT RESULTS AND DISCUSSIONS

The accuracy experiment is divided into 1) accuracy class of the new EVT and 2) accuracy class of the EVT under driving the AMI smart meter. Finally, the two experimental results are compared to evaluate the feasibility of the new EVT to replace the traditional VT for AMI smart meter. Considering the measurement errors of the two voltage differential probes and the sampling errors of the NI DAQ system, this study first evaluates and calibrates the differences between the two combinations. Each combination means one voltage differential probe with one input channel of the NI DAQ system. The AC power supply synchronously sends 114 V or 380 V to the two voltage differential probes with the NI DAQ system. Then voltage and phase differences between the two combinations are calculated by a LabVIEW-based measurement platform, and the results are listed in Table II. In 80 % to 120 % of the rated voltage (110 V or 380 V), the voltage differences are all less than 0.015 %, and the phase differences are all less than 0.25'. Therefore, the measurement results prove that voltage and phase differences between the two standard VTs is extremely small. Finally, these differences are corrected in subsequent experiments.

TABLE II RATIO DIFFERENCE AND PHASE DIFFERENCE BETWEEN TWO REFERENCE VTs (INCLUDING NI DAQ SYSTEM)

Voltage level		Voltage difference (%)	Phase difference (')
114 V	80 %	0.0147	0.107
	90 %	0.0088	0.055
	100 %	0.0082	-0.099
	110 %	0.0111	0.041
	120 %	0.0071	0.224
380 V	80 %	-0.0081	-0.075
	90 %	-0.0083	-0.084
	100 %	-0.0075	-0.054
	110 %	-0.0060	-0.065
	120 %	-0.0087	-0.204

A. Accuracy class test for new EVT without AMI smart meters

In this subsection, we conduct an accuracy class test for the new EVT according to IEC Standard 60044-7. At 80 %, 90 %, 100 %, 110 %, and 120 % of rated voltage (380 V), voltage errors and phase displacements of the new EVT are measured. The measurement results are shown in Table 3. Overall, the voltage errors are all less than 0.02 % and the phase displacements are all less than 0.3'. Therefore, the new EVT can meet accuracy Class 0.1, which has been defined in IEC Standard 60044-7.

TABLE III RATIO ERROR AND PHASE DISPLACEMENT FOR NEW EVT WITHOUT AMI SMART METERS

Percentage of rated voltage (%)	Voltage (ratio) error ε_u (%)	Phase displacement $\Delta\varphi$ (')
80	0.0152	0.285
90	0.0044	0.012
100	-0.0001	0.000
110	0.0073	0.029
120	-0.0057	-0.205

B. Accuracy class test for new EVT with an AMI smart meter

After the new EVT output is connected to an AMI smart meter, we re-test the accuracy class for the EVT, and simultaneously test whether the AMI smart meter is driven and works normally. The test results are shown in Fig. 9, the AMI smart meter is successfully driven by the new EVT, and the panel of the AMI meter displays 114.00 V. At this time, the test voltage is 380 V. However, since the voltage ratio setting of the VT in the AMI meter is still the default value (1:1), the panel of the AMI meter directly displays the secondary voltage value of the new EVT, which is 114.00 V. Besides, the measurement results for accuracy class are listed in Table IV. Overall, the voltage errors are all less than 0.04 % and the phase displacements are all less than 0.1'. Therefore, the new EVT still meets accuracy Class 0.1.

From the above two test results, it can be seen that when the new EVT drives the AMI smart meter, the voltage error slightly increases. Especially at 120 % of the rated voltage, the voltage error is the most serious. Therefore, for 120 % of the rated voltage, this study analyzes the voltage waveform of the new EVT with or without driving the AMI smart meter. The measurement results of voltage waveforms are shown in Figs. 10 and 11. When driving an AMI smart meter, the noise of the new EVT waveform is slightly increased due to the high-power output of the push-pull power amplifier. In contrast, the noise in the new EVT waveform is slightly reduced when the AMI smart meter is not present. However, the new EVT satisfies Class 0.1 whether or not it drives an AMI smart meter.

TABLE IV RATIO ERROR AND PHASE DISPLACEMENT FOR NEW EVT WITH AN AMI SMART METER

Percentage of rated voltage (%)	Voltage (ratio) error ε_u (%)	Phase displacement $\Delta\varphi$ (')
80	0.0167	-0.016
90	0.0041	0.074
100	-0.0002	0.003
110	0.0037	0.071
120	-0.0358	-0.006

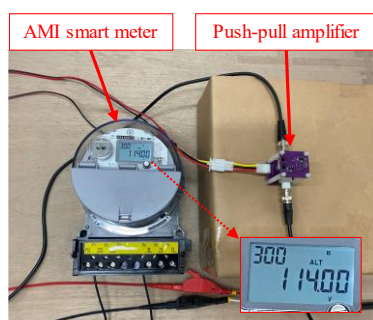


Fig. 9. Display panel of the AMI smart meter powered by the new EVT.

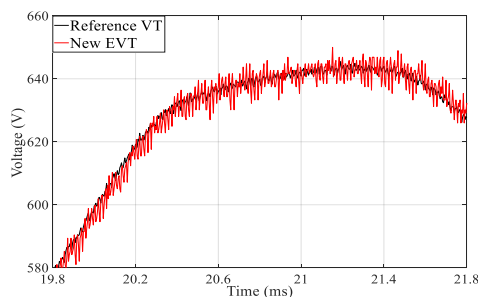


Fig. 10. Waveform at 120 % of the rated voltage (without AMI smart meters).

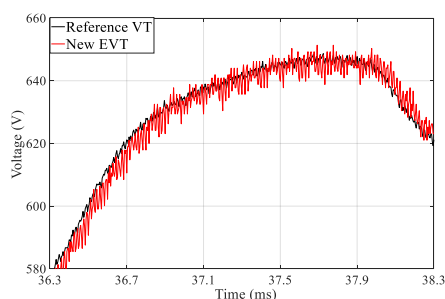


Fig. 11. Waveform at 120 % of the rated voltage (with the AMI smart meter).

V. CONCLUSION

This study completed a new EVT applied to AMI smart meters. The new EVT is mainly based on a resistive voltage divider combined with a push-pull power amplifier, which can realize 380 V/114 V voltage measurement and drive AMI smart

meters. In addition, the new EVT adopts a modular design. Therefore, according to the application of different distribution voltage levels, the voltage measurement requirements of 11.4 kV/114 V or 22.8 kV/114 V can be realized by cascading modules. The experimental results show that when the new EVT successfully drives the AMI smart meter, its measurement accuracy can still meet Class 0.1 according to IEC standard 60044-7.

REFERENCES

- [1] M. D. L. del Casale, "On multistress aging of epoxy resins: PD and temperature," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 8, no. 2, pp. 299–303, Apr. 2001.
- [2] H. Wang, Y. Gu, and X. Cheng, "Electrical aging experiment of epoxy resin insulation equipment and research on the influence of electrical field uniformity," in *Proc. 2017 4th International Conference on Electric Power Equipment - Switching Technology (ICEPE-ST)*, Xi'an, China, Oct. 2017.
- [3] M. Saitoh, T. Kimura, Y. Minami, N. Yamanaka, S. Maruyama, T. Nakajima, and M. Kosakada, "Electronic instrument transformers for integrated substation systems," in *Proc. IEEE/PES Transmiss. Distrib. Conf. Exhib.*, Yokohama, Japan, Oct. 2002.
- [4] Marco Faifer, Sergio Toscani, Roberto Ottoboni, "Electronic combined transformer for power-quality measurements in high-voltage systems," *IEEE Trans. Instrum. Meas.*, vol. 60, no. 6, pp. 2007–2013, 2011.
- [5] S. Kucuksari and G. G. Karady, "Experimental comparison of conventional and optical VTs, and circuit model for optical VT," *IEEE Trans. Power Del.*, vol. 26, no. 3, pp. 1571–1578, Jul. 2011.
- [6] G. Crotti, D. Gallo, D. Giordano, C. Landi, and M. Luiso, "Medium voltage divider coupled with an analog optical transmission system," *IEEE Trans. Instrum. Meas.*, vol. 63, no. 10, pp. 2349–2357, Oct. 2014.
- [7] B. Li, Z. Zhang, and L. Ding, "Self-detecting the measurement error of electronic voltage transformer based on principal component analysis-wavelet packet decomposition," *IEEE Trans. Instrum. Meas.*, vol. 70, Oct. 2021, art. no. 3526609.
- [8] *Instrument Transformers-Part 7: Electronic Voltage Transformers*, IEC Standard 60044-7, Dec. 1999.
- [9] R. C. Allil and M. M. Werneck, "Optical high-voltage sensor based on fiber Bragg grating and PZT piezoelectric ceramics," *IEEE Trans. Instrum. Meas.*, vol. 60, no. 6, pp. 2118–2125, Jun. 2011.
- [10] J. C. Santos, M. C. Taplamacioglu, and K. Hidaka, "Pockels high-voltage measurement system," *IEEE Trans. Power Del.*, vol. 15, no. 1, pp. 8–13, Jan. 2000.
- [11] H. Liu, F. Zhou, L. Chen, M. Lei, X. Yin, C. Jiang, and J. Liu, "The development of precision 500/ $\sqrt{3}$ -kV two-stage voltage transformer with high-voltage excitation," *IEEE Trans. Instrum. Meas.*, vol. 70, Jan. 2021, art. no. 1004407.
- [12] M. Zucca, M. Modarres, D. Giordano and G. Crotti, "Accurate numerical modelling of MV and HV resistive dividers," *IEEE Trans. Power Del.*, vol. 32, no. 3, pp. 1645–1652, Jun. 2017.